

ZAHIDUL HOQUE

Aspiring Ph.D. Researcher in AI, Computer Vision & Autonomous Systems

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🌐 [linkedin.com/in/zahidul-hoque](https://www.linkedin.com/in/zahidul-hoque) 🐙 github.com/ZAHIDUL-HOQUE 📁 Thesis Code Repository

Research Interests

Core Domains: Computer Vision & Deep Learning, Connected and Autonomous Vehicles (CAV), LiDAR & Multimodal Sensor Perception, Deep Metric Learning, Digital Twins & Sim2Real Perception, Vision-Language-Action (VLA) Models, Traffic Safety & Vulnerable Road User (VRU) Protection, Long-Tailed & Class-Imbalanced Learning.

Education

Master of Science (M.Sc.) in Data Science and Analytics with Advanced Research Sep 2024 – Present

University of Hertfordshire

Hatfield, United Kingdom

M.Sc. Thesis Project: *Prototypical Mask R-CNN for Vehicle Exterior Damage Detection* (Supervisor: Dr. Joseph Reddington).

Focus Areas: Advanced Computer Vision, Deep Learning Architectures, Metric Learning, High-Dimensional Feature Space Geometry, Statistical Machine Learning, Neural Networks.

Bachelor of Science (B.Sc.) in Computer Science

Sep 2019 – Jun 2023

Beijing Institute of Technology (BIT)

Beijing, China

Core Curriculum: Data Structures & Algorithms, Linear Algebra, Multivariable Calculus, Probability & Mathematical Statistics, Operating Systems, Computer Architecture, Software Engineering.

Master's Research & Technical Contributions

Prototypical Mask R-CNN for Vehicle Exterior Damage Detection

Sep 2024 – 2025

Master's Final Research Project (7COM1039) | University of Hertfordshire | Supervisor: Dr. Joseph Reddington

Novel Deep Learning Architecture: Designed and implemented a hybrid instance segmentation architecture integrating **Mask R-CNN** with **Deep Metric Learning**, replacing the standard parametric linear classification layer with a **Squared Euclidean Prototypical Predictor head**.

Tackling Extreme Class Imbalance: Formulated a geometric centroid-learning framework to resolve the severe long-tail distribution and gradient dominance inherent in vehicle exterior inspection datasets (CarDD benchmark), eliminating minority class collapse (e.g., flat tires, shattered glass).

Hyperspherical Latent Geometry: Enforced strict L_2 normalisation on both 1024-D RoI feature embeddings and learnable class prototypes, projecting representations onto a unit hypersphere ($\|p\| = 1$). Derived mathematically that the pairwise squared Euclidean distance $d^2 = 2 - 2\cos(\theta)$ achieved empirical values of $1.91 \leq d^2 \leq 2.09$ ($\theta \approx 90^\circ$), proving that the model converged to an **orthogonal feature space** that eliminates inter-class visual confusion.

Optimization & Training Protocol: Leveraged a ResNet-50-FPN backbone pre-trained on MS-COCO, configured with temperature-scaled softmax ($\alpha = 20$) for rapid monotonic loss convergence, SGD with momentum, and learning rate step-decay. Utilised Automatic Mixed Precision (`torch.cuda.amp`) to optimize memory on an NVIDIA RTX 3060 (12GB).

Data Augmentation & Consistency: Engineered synchronized spatial transformations (`torchvision.transforms`) dynamically mirroring binary segmentation masks and bounding box coordinates during horizontal flips to maintain pixel-level spatial fidelity under varying sensor orientations.

Empirical Results: Validated across 810 test images, attaining a **Mean Average Precision (mAP) of 0.319** (IoU 0.50:0.95), **AP@50 of 0.630**, and an **Average Recall (AR) of 0.436**, demonstrating robust minority class retrieval.

Future Vision-Language-Action (VLA) Roadmap: Proposed extending metric damage embeddings into the latent space of Multimodal Large Language Models (MLLMs) and VLA frameworks for automated, explainable vehicle safety diagnostics in autonomous mobility ecosystems.

Open Science: Fully documented and published the complete modular PyTorch codebase and training logs on GitHub: [GitHub Repository Link](#).

Alignment with UrbanITY Lab & Research Statement

LiDAR, Camera Perception & Sensor Fusion: Experience with multi-scale feature hierarchies (FPN), RoIAlign, and metric latent spaces directly translates to roadside and vehicle-mounted 3D LiDAR/Camera object detection pipelines (e.g., DINOSTAR, LiGuard) developed at UrbanITY Lab.

Digital Twins & Sim2Real Gap (UrbanTwin): Strong mathematical foundation in geometric regularisation, domain adaptation, and feature normalization provides an ideal basis for tackling domain shifts between synthetic digital-twin LiDAR datasets (LUMPI, V2X-Real-IC, TUMTraf-I) and real-world sensor streams.

Traffic Safety & VRU Protection: Passionate about applying real-time deep learning to vulnerable road user (VRU) detection, smart work zone monitoring, and incident mitigation for connected and automated transportation systems.

Technical Skills

Programming Languages: Python (Proficient), C/C++, SQL, Bash/Shell Scripting.

Machine Learning & CV: PyTorch, Torchvision, Pycocotools, OpenCV, Mask R-CNN, Faster R-CNN, Feature Pyramid Networks (FPN), Deep Metric Learning, Prototypical Networks, Scikit-learn.

Data Science & Visualization: NumPy, Pandas, SciPy, Matplotlib, Seaborn, PIL.

Mathematical Foundations: Multivariable Calculus, Linear Algebra, High-Dimensional Vector Geometry, Probability & Statistics, Optimization Algorithms.

Developer Tools & Infrastructure: Git/GitHub, Linux/Ubuntu environments, NVIDIA CUDA, Mixed Precision Training (AMP), Jupyter Notebook, LaTeX (Overleaf), Figma.

Work & Project Experience

Project Associate (Computer Vision & Data Annotation)

Feb 2023 – Mar 2023

Quantanite

Dhaka, Bangladesh

Executed high-precision data annotation, bounding box generation, and segmentation mask labeling for client machine learning pipelines.

Conducted systematic data cleaning and quality auditing to eliminate label noise, enhancing downstream model training convergence.

Certifications & Training

ML in Production: From Data Scientist to ML Engineer – Practical model deployment, monitoring, and pipeline orchestration.

Python Programming: Practical & Real-Time Systems – Advanced object-oriented programming, data structures, and algorithmic optimization.

Professional Back-Office Services – Skills for Employment Investment Program (SEIP), Ministry of Finance, Bangladesh.

Academic References

Dr. Joseph Reddington

Senior Lecturer / Academic Supervisor, School of Physics, Engineering and Computer Science

University of Hertfordshire, Hatfield, United Kingdom

Supervised Master's Thesis: "Prototypical Mask R-CNN for Vehicle Exterior Damage Detection"

Email: Available upon request

Academic Profile: researchprofiles.herts.ac.uk